

An LLM-based Cross-Domain Fault Localization in Carrier Networks

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Abstract—The increasing complexity of carrier networks poses significant challenges for root cause analysis (RCA) and fault localization across diverse domains, including Optical Transport Networks (OTN) and IP Radio Access Networks (IPRAN). Traditional approaches, predominantly reliant on rule-based systems and manual expertise, are often inadequate to manage the scale and intricacies of modern network environments. In this paper, an intelligent method, CrossRCA, is presented to address cross-domain fault localization and RCA in carrier networks. Powered by large language models (LLMs), CrossRCA is designed to dynamically align incoming alerts with relevant diagnostic workflows, aggregate critical context-aware diagnostic information, predict incident root cause categories, and generate explanatory narratives to support network engineers. Pre-trained LLMs, augmented with prompt engineering and Retrieval Augmented Generation (RAG) techniques, are utilized to encapsulate domain-specific knowledge and enable sophisticated contextual reasoning across heterogeneous network domains. As a result, accurate and automated fault diagnosis is facilitated. Experimental evaluations conducted on real-world network fault datasets demonstrate significant improvements in diagnostic accuracy and reductions in troubleshooting time, particularly in complex cross-domain scenarios. The transformative potential of LLM-driven frameworks to advance autonomous network management and enhance operational efficiency in carrier networks is highlighted in this paper.

Keywords—root cause analysis; large language models; fault localization

I. INTRODUCTION

The rapid evolution of carrier networks, driven by increasing demands for high-speed and reliable communication, has led to the development of complex network infrastructures such as Optical Transport Networks (OTN) and IP Radio Access Networks (IPRAN). OTN delivers high-capacity, low-latency, and highly reliable transport solutions, forming the backbone of modern communication networks [1]. Meanwhile, IPRAN plays a critical role in mobile backhaul, connecting base stations to the core network and enabling seamless communication across heterogeneous network environments [2]. While these technologies are fundamental to carrier network infrastructure, their interdependencies pose significant challenges in network management, particularly in fault localization and root cause

analysis (RCA) [3]. In the domain of network fault localization, significant challenges for automated RCA are presented by the hierarchical and domain-specific architecture of carrier networks. These networks are typically structured into multiple layers and specialized domains, such as OTN and IPRAN, which are interdependent. Faults originating in lower layers can propagate upward, leading to cascading failures across domains. For instance, a failure in an OTN layer may result in widespread node malfunctions in the corresponding IPRAN layer, subsequently leading to large-scale base station outages [4]. The IPRAN provides critical backhaul connectivity for 4G and 5G base stations. Due to stringent network security and reliability requirements, key connections between IPRAN nodes are frequently supported by the OTN. Faults or performance degradation in OTN or IPRAN can result in significant disruptions to mobile services that rely on these networks. To address these challenges, IPRAN and OTN, two highly interdependent network domains, are focused on in this paper due to their susceptibility to frequent failures.

Fault localization and RCA are critical for ensuring network reliability. By accurately identifying the underlying issues and preventing their recurrence, RCA not only restores service availability swiftly but also strengthens the overall resilience of communication networks [3]. However, traditional methods, which predominantly rely on rule-based systems and manual expertise, face significant limitations in complex network environments. These methods involve predefined rules and static thresholds to identify potential issues, requiring domain experts to manually interpret data and locate root causes. While effective in simple and small-scale networks, such approaches are increasingly inadequate in addressing the intricacies of contemporary carrier networks [5]. Modern networks are characterized by large-scale infrastructures, dynamic behaviors, and cross-domain dependencies. Traditional methods are ill-equipped to handle these interdependencies, leading to prolonged troubleshooting times, inefficiencies, and escalated operational costs. Significant challenges are also faced by traditional methods due to the diverse data availability, which ranges from excessive logs and alarms to sparse, inconsistent, and noisy information. These issues hinder pattern recognition and accurate diagnosis. Additionally, reliance on static troubleshooting guides, often outdated and unsuitable for novel

or multi-causal incidents, further delays resolution [6]. As cloud and software-defined networks scale, static, manual approaches are increasingly inadequate. These limitations highlight the urgent need for intelligent, automated methods to efficiently analyze heterogeneous data, adapt to dynamic conditions, and improve network reliability.

To address the limitations of traditional methods, AI-driven approaches have been increasingly adopted in network management [7-9]. Techniques such as machine learning and deep learning have shown promise in automating fault detection and RCA by analyzing large volumes of network data, including logs, performance metrics, and telemetry [10]. Patterns and anomalies that are often invisible to human operators can be identified by these methods, enabling faster and more accurate fault localization. However, AI-based methods typically rely on structured data and require extensive feature engineering, which limits their adaptability in highly dynamic and heterogeneous network environments like OTN and IPRAN. Additionally, the ability to incorporate contextual and domain-specific knowledge, which is critical for cross-domain fault localization and RCA, is often lacking in these approaches.

New avenues for automating network management tasks have been opened by recent advancements in LLMs. LLMs, which are pre-trained on massive datasets, excel in natural language understanding, contextual reasoning, and knowledge encapsulation [11]. A transformative capability for processing high-volume data, extracting pertinent information, and generating concise, contextually relevant outputs is offered by LLMs [12]. By automating the identification and interpretation of critical data, LLMs substantially reduce the operational burden on communication engineers, who would otherwise need to manually analyze extensive datasets. This enables engineers to prioritize incident resolution with greater efficiency and precision, thereby improving the overall effectiveness of the troubleshooting process. Unlike traditional AI models, LLMs can process unstructured data, such as logs and alerts, and generate human-readable narratives that provide actionable insights. By leveraging LLMs, network management systems can integrate contextual and cross-domain reasoning capabilities, enabling more sophisticated and automated fault localization and RCA.

In this paper, an intelligent method leveraging LLMs is proposed to address the challenges of cross-domain fault localization and RCA in carrier networks, based on real-world network fault datasets. The proposed system achieves a substantial improvement in the rapid localization of cross-domain faults, thereby enhancing operational efficiency and ensuring service reliability.

The structure of the paper is organized as follows: Section II introduces the key technologies underlying the development of CrossRCA, including prompt engineering and RAG. Section III provides a detailed explanation of the methodology, covering the overall framework of CrossRCA, data preprocessing, data analysis, and the application of LLMs in constructing the system. The implementation details and experimental results are presented and discussed in Section IV. Finally, conclusions are drawn in Section V.

II. KEY TECHNOLOGIES FOR BUILDING THE CROSSRCA

The implementation of CrossRCA incorporates state-of-the-art technologies to achieve robust and accurate RCA in carrier networks. Key components, such as prompt engineering and RAG, serve as the cornerstone of the CrossRCA intelligent method. Each of the technologies plays a pivotal role in addressing the scale, complexity, and dynamic nature of modern carrier networks.

A. Prompt Engineering

Prompt engineering is recognized as a critical technique for tailoring LLMs to domain-specific tasks. This strategic approach involves the design of task-specific instructions, referred to as prompts, to enhance the accuracy, relevance, and context-sensitivity of outputs generated by LLMs for downstream applications [13]. A broad spectrum of methodologies is encompassed within this field, ranging from fundamental techniques such as zero-shot and few-shot prompting to advanced methods like chain-of-thought (CoT) prompting [14-15]. In CrossRCA, prompt engineering is employed to guide the LLM in processing complex and contextual information. Domain-specific knowledge is encapsulated within carefully designed prompts to ensure the generation of contextually relevant and actionable outputs. This approach enables the LLMs to dynamically adapt to varying fault scenarios, facilitating precise alignment between incoming alerts and diagnostic workflows. By optimizing prompt formulations, the reasoning and comprehension capabilities of the LLM are maximized, thereby enhancing its effectiveness in cross-domain fault localization and RCA.

B. Retrieval Augmented Generation (RAG)

Remarkable capabilities are demonstrated by LLMs, yet they are faced with inherent challenges such as the generation of hallucinated information, reliance on outdated knowledge, and the exhibition of biases in decision-making processes [16]. A promising approach to addressing these limitations is RAG, which integrates external knowledge bases into the model's reasoning process. By dynamically retrieving and incorporating domain-specific information from external databases, RAG effectively bridges the gap between the general pre-trained knowledge of LLMs and the specific contextual requirements of specialized tasks [17]. This integration enhances the reliability, accuracy, and relevance of the generated outputs, offering a robust solution to overcome the limitations of standalone LLMs in complex, domain-sensitive applications [18]. The integration of RAG within CrossRCA addresses the challenge of retrieving relevant context from vast and heterogeneous knowledge bases, such as historical fault data, troubleshooting guides, and network topologies [19]. By enabling the real-time retrieval of pertinent information from external data sources, RAG enhances the capabilities of the LLM, ensuring that response generation is informed by accurate and contextually relevant knowledge. This approach allows diverse data formats to be effectively handled, while reasoning capabilities are strengthened across multiple network domains. Furthermore, the use of RAG ensures consistency with the most current and reliable information, which is essential in the dynamic and rapidly evolving carrier network environment.

C. Synergistic Integration of Key Technologies

The limitations of traditional RCA methods are effectively addressed in CrossRCA through the combination of prompt engineering and RAG [3]. Prompt engineering is employed to ensure that the LLM accurately interprets and responds to the domain-specific complexities of carrier networks, while RAG provides access to a rich and dynamic knowledge base. By integrating these technologies, sophisticated contextual reasoning is enabled, fault localization and RCA are performed more efficiently, and actionable insights are delivered. This approach ultimately reduces troubleshooting time and enhances operational efficiency in carrier networks.

III. CROSS-DOMAIN FAULT LOCALIZATION AND ROOT CAUSE ANALYSIS

The architecture of the LLM-based cross-domain intelligent network operation and maintenance system, as illustrated in Figure 1, primarily comprises a data collection module and the NetRCA module. The data collection module is responsible for acquiring data relevant to fault localization. Alarm data are obtained through the operator's Enterprise Service Bus (ESB), which facilitates the collection of alarm data from multi-vendor network devices, including IPRAN and OTN network elements. Topology information is retrieved from the operator's resource management system. Both alarm data and topology information are gathered via Kafka interfaces, enabling their integration and utilization by the IPRAN and OTN fault localization modules. This integration supports single-domain fault localization as well as cross-domain fault localization for IPRAN and OTN networks. The NetRCA (Network Root Cause Analysis) module forms the core of the fault localization and RCA system, leveraging LLMs to enhance analysis capabilities across diverse domains. This module comprises several key components, including data preprocessing, embedding generation, and LLM-based reasoning enhanced through RAG and prompt engineering techniques.

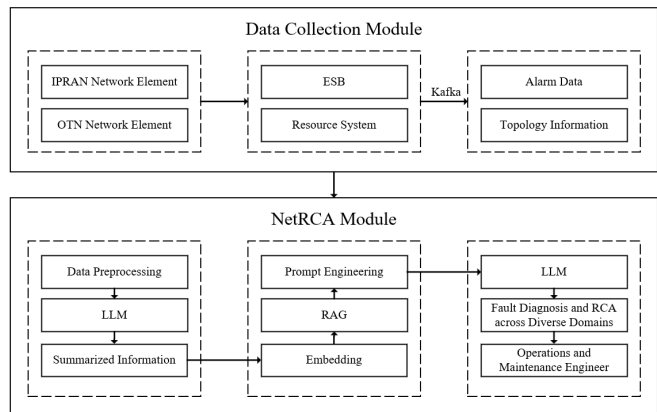


Figure 1: Overall framework of CrossRCA

A. Data Analysis

A comprehensive statistical analysis was performed on the collected data. The daily fault occurrences in the OTN and IPRAN networks were initially examined, with the findings presented in Figure 2. The fault occurrence trends in both networks exhibit a high degree of similarity, indicating a strong interdependence between the OTN and IPRAN domains. This

observation highlights the critical importance of conducting integrated analysis of these closely interdependent network systems.

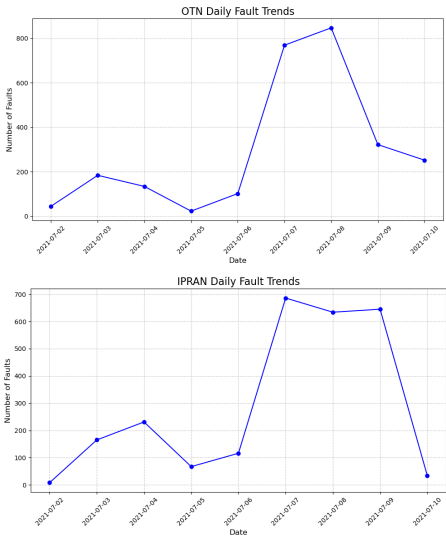


Figure 2: Comparison of Alarm Trends in IPRAN and OTN

Subsequently, the proportions of alarm names for OTN and IPRAN networks were analyzed, as depicted in Figure 3. The results reveal distinct patterns in alarm distributions. A significant proportion of alarms in OTN networks, such as MUT_LOS and R_LOS, highlights the pressing need to address reliability issues at the transport layer. Similarly, in IPRAN networks, alarms including PW_VC Status Down and Physical Port Down dominate, underscoring the critical focus required on link-layer stability. These observations underscore the necessity of implementing fault diagnosis and troubleshooting strategies tailored to the distinct characteristics of each network. Moreover, the importance of developing a cross-domain intelligent operation and maintenance system is validated by these findings, as such a system is essential for addressing the interdependencies and fault scenarios inherent in both OTN and IPRAN networks.

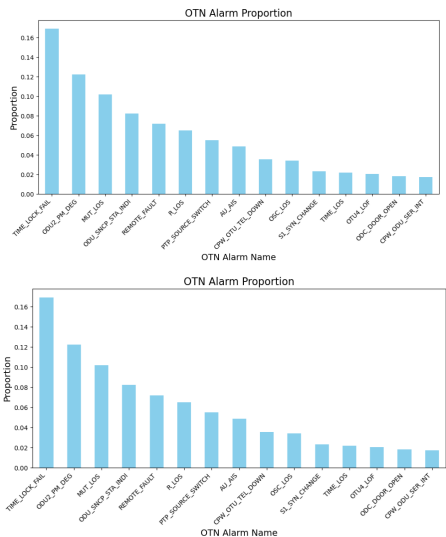


Figure 3: Analysis of Alarm Name Proportions in IPRAN and OTN

B. Data Preprocessing

The data preprocessing framework is specifically designed to process and refine raw alarm data, which frequently contains extraneous and irrelevant information. As illustrated in Figure 4, the preprocessing pipeline incorporates several critical steps applicable to both IPRAN and OTN networks, ultimately leading to temporal correlation analysis for fault localization. A detailed explanation of each step is provided below.

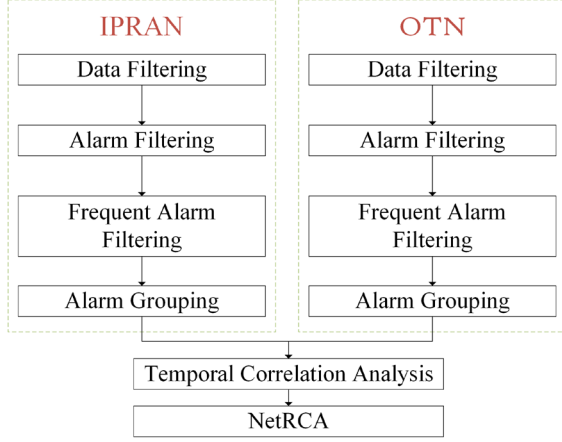


Figure 4: Alarm data preprocessing workflow for IPRAN and OTN

Data filtering: Both IPRAN and OTN alarms are filtered based on their occurrence time. Only alarms occurring within the same time frame are retained, as alarms with significant time differences are unlikely to exhibit correlations. Define the set of alarms for IPRAN and OTN as A_{IPRAN} and A_{OTN} , respectively. For a given time frame T , retain alarms where:

$$A_{filtered} = \{a \in A_{IPRAN} \cup A_{OTN} \mid t_a \in T\} \quad (1)$$

where t_a is the occurrence time of alarm a .

Alarm filtering: For IPRAN and OTN networks, alarms are further cleaned to remove entries that lack essential information. Specific to IPRAN, alarms at the aggregation layer with high correlation to OTN alarms are prioritized.

Frequent alarm filtering: To refine the dataset, frequent alarms are cleaned based on the following rules: For repeated alarms $a_1, a_2, \dots, a_n$ on the same port within $\Delta t_1 = 1$ minute:

$$A_{reduced} = \{a_1\} \quad (2)$$

For alarms with duration d_a less than 1 minute and appearing once within $\Delta t_2 = 5$ minutes:

$$A_{cleaned} = \{a \in A_{reduced} \mid d_a > 1 \text{ min or } f_a > 1\} \quad (3)$$

where f_a is the frequency of a .

Alarm grouping: After filtering, alarms are grouped into fixed time intervals for subsequent analysis. The Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm is applied to perform this grouping, ensuring alarms are categorized based on temporal proximity. As shown in Equation 4, G_i represents a group.

$$G = \{G_1, G_2, \dots, G_n \mid G_i = \{a \in A_{cleaned}\}\} \quad (4)$$

Temporal correlation analysis: After grouping alarms using DBSCAN, temporal correlation analysis is performed to identify potential cross-network dependencies:

Single-Fault detection: If an IPRAN alarm group G_i^{IPRAN} contains only one alarm, it is classified as an IPRAN-specific fault.

Cross-Network correlation: For IPRAN alarm groups with multiple alarms G_i^{IPRAN} , the first alarm's timestamp t_{first} is compared to that of OTN alarm groups G_j^{OTN} . If the timestamp $|t_{first}^{IPRAN} - t_{first}^{OTN}| \leq 60$ seconds, the alarms are deemed correlated.

Data extraction and conversion: Key data from correlated IPRAN and OTN groups (G_i^{IPRAN}, G_j^{OTN}), are extracted and transformed into textual format for analysis by LLMs. The final data text is shown in the figure 5. By converting the raw data into textual format, the framework leverages the natural language processing capabilities of the LLMs to achieve more accurate fault localization and actionable insights.

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IPRAN Alarms:
Network Element Name, Alarm Name, Occurrence Time,
Location Information
HJ-FWLYYY-2010-FWUH(FK-HJ), Physical Port Down,2021-07-
09 23:21:38,Physical Entity Name=GigabitEthernet3/0/6 Fault
Cause=Physical Port Down Port Alias=To-GDZQ-fengwuGuYu-
1F-980C-SM-A1-FK06-GigabitEthernet0/5/0
.....
HJ-FWLYYY-2010-FWUH(FK-HJ), ETH_LOS, 2021-07-09
23:21:53, Port Name=GigabitEthernet3/0/4 Slot Number=3 Card
Number=0 Port Number=4 Port Alias=d1:FK04-fengwuLongGang-
2.0-10GE0/5/0
OTN Alarms:
Alarm Name, Alarm Source, Occurrence Time, Location
Information
OLP_PS,KSZSYXX / HJ-H KSZSYXX-8377-SD-OTN,2021-07-09
23:21:34,3-LYYY direction Even Wave-6-13 OLP-1(ZSCGH
Main direction)-OTS.1
OLP_PS,KSZSYXX / HJ-H KSZSYXX-8374-SD-OTN,2021-07-
09 23:21:34,1-ZSYXX direction Odd Wave (Backup)-4-13OLP-
1(ZSCGH Main direction)-OTS.1
.....
MUT_LOS,KS-FWLYYY / HJ-H KS-FWLYYY-8374-SD-
OTN,2021-07-09 23:22:24,0-ZSYXX direction Main direction-12-
52DAPXF-11(LIN/LOUT)-OTS.1
  
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Figure 5: Alarm Data Examples for IPRAN and OTN

C. Fault Localization and RCA Based on LLMs

After data preprocessing, the alarm data often exceeds manageable tokenized input lengths, complicating the reasoning process for identifying incident root causes. Alarm information, as depicted in Figure 5, frequently surpasses 2,000 tokens and exhibits low readability, posing significant challenges for direct processing by LLMs. The excessive length not only exceeds the input capacity of many LLMs but also introduces noise, further complicating prediction tasks. While LLMs have shown potential in automatic summarization, their performance on domain-specific or long-tail tasks remains suboptimal without appropriate guidance. Consequently, directly feeding such verbose data into an LLM for fault localization often results in suboptimal outcomes. To address this issue, an additional preprocessing layer is introduced, leveraging the summarization capabilities of LLMs to condense alarm information prior to diagnostic reasoning. The summarization prompt, carefully designed as shown in Figure 6, ensures that the LLM focuses exclusively on extracting critical information while omitting irrelevant details. This approach effectively reduces token length and enhances both

the interpretability and utility of the alarm data, thereby improving its suitability for subsequent fault localization tasks.

You are a professional network operations engineer with extensive experience in network maintenance and management. Please summarize the above input. Please note that the above input is IPRAN and OTN alarm data. The summary results should be concise and accurate, and should cover important information as much as possible. Just return the summary without any additional output.

Figure 6: Prompts of summarizing Alarm Data

Due to the proprietary and sensitive nature of alarm data, LLMs are unable to access this domain-specific knowledge during their pretraining phase. Consequently, LLMs may fail to comprehend the specialized terminology, structure, and contextual relationships within the alarm data or to identify appropriate solutions for addressing these alarms. This limitation poses significant challenges for fault localization and RCA in complex network environments.

To overcome this constraint, it is essential to supplement the alarm data with relevant proprietary knowledge when feeding inputs to the LLM. By leveraging RAG and prompt engineering techniques, domain-specific resources—such as troubleshooting guides, alarm descriptions, historical resolution records, and contextual alarm explanations—can be retrieved and incorporated into the input prompt. This enriched input provides the LLM with the necessary background knowledge to interpret alarm data accurately, identify root causes, and propose effective solutions. RAG dynamically retrieves the most relevant domain knowledge (e.g., troubleshooting documents or prior cases) based on the alarm data, ensuring the model has access to updated and case-specific information. Prompt Engineering structures and integrates both the alarm data and retrieved knowledge into a coherent input format that aligns with the LLM's natural language processing capabilities.

By integrating these approaches, a deeper understanding of proprietary alarm data is achieved by the LLM, enabling it to: detect subtle patterns and dependencies within alarms; correlate alarms across diverse network domains (e.g., IPRAN and OTN); produce accurate, context-aware fault diagnosis; and recommend actionable, scenario-specific solutions. This methodology significantly enhances the performance of LLMs in fault localization and root cause analysis, effectively bridging the gap between general language model pretraining and domain-specific requirements.

IV. IMPLEMENTATION AND RESULTS

The dataset employed in this study consists of historical alarm data and topology connection data collected over a specific period from IPRAN and OTN networks in Guangdong Province. The historical alarm data include 2,589 records from IPRAN and 8,555 records from OTN. Key alarm information encompasses network element names, alarm names, timestamps, and location details, as illustrated in Figure 5.

A. Traditional methods

Traditional manual methods for fault correlation are time-intensive and error-prone. For example, upon receiving an IPRAN-specific fault alarm, maintenance personnel typically consult OTN teams via phone, cross-check network records, and inspect management systems to identify whether the fault

originates within the IPRAN domain. This process often takes 10 minutes per case and can extend further during high-concurrency fault scenarios. Additionally, manual processes are susceptible to human error, leading to potential misjudgments in fault localization. Expert systems employing predefined rules offer some degree of automation but lack the flexibility to adapt to evolving network topologies and fault scenarios. These systems are often unable to handle complex network dynamics, making them less effective in real-world applications.

B. Results and Discussion

Experimental evaluations conducted on real-world network fault datasets demonstrate outstanding performance in cross-domain fault localization. As shown in Figure 7, the actual cross-domain network topology of IPRAN and OTN is illustrated, with dashed lines indicating the actual fault points. Figure 8 presents the fault localization results predicted by Cross-RCA. A comparison between the actual fault data and the model-generated diagnoses clearly shows that the proposed Cross-RCA system effectively identifies fault points in real-world scenarios.

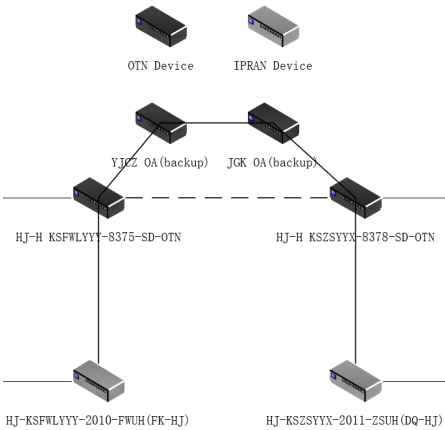


Figure 7: The actual topology of the alarm data, with dashed lines indicating the actual fault points.

Link Issues Reflected by Alarms
Based on the alarm information:
The physical port down alarms on the IPRAN (GigabitEthernet3/0/6 and 3/0/4) are strongly correlated with the primary fiber issues on the OTN (OSC_LOS, OLP_PS).
The issue is concentrated on the primary link, possibly due to fiber breakage or severe attenuation.

Preliminary Fault Localization
Primary Link:
The fault is concentrated on the primary fiber segment between "HJ-H KSWLYYY-8375-SD-OTN" and "HJ-H KSZSYXX-8378-SD-OTN."
This link carries traffic for both IPRAN GigabitEthernet3/0/6 and 3/0/4. A fiber break will result in synchronization loss for upstream and downstream devices.
2. Backup Link:
The backup link did not trigger a switch either, which may indicate that the OLP module is malfunctioning or the backup link also suffers from quality issues.

Root Causes of the Fault
1. Fiber Breakage or Severe Attenuation:
The primary fiber segment connecting "HJ-H KSWLYYY" and "HJ-KSZSYXX" is interrupted.
2. OLP Switching Anomaly:
The backup link failed to take over the traffic, likely due to OLP module failure or substandard backup link quality.

Fault Localization Steps and Verification Methods
1. Fiber Testing:
Use OTDR to test from both ends of the fiber ("HJ-H KSWLYYY" and "HJ-KSZSYXX") to locate breakpoints or points of abnormal attenuation.
2. OLP Inspection:
Check the OLP switching logs to determine if a backup link switch was attempted.
Verify the quality of the backup link.
3. Optical Module Replacement:
Replace the optical modules involved in the affected links to rule out hardware issues.

Fault Repair Measures
1. Repair the primary fiber link.
2. Replace faulty optical modules or relay equipment (e.g., OA, OLP).
3. Test the quality of the backup link to ensure OLP switching functionality.

Figure 8: Cross-domain root cause analysis and fault localization results obtained from CrossRCA.

This system significantly bolsters the capabilities for cross-layer fault localization, thereby reducing the time required for manual troubleshooting and enhancing operational and

maintenance efficiency. It enables cross-domain network fault localization within seconds, marking a substantial improvement in efficiency compared to the traditional minute-level localization methods, therefore, the manual work time was reduced by more than 90%. By providing specific solutions for identified faults, the system not only expedites the resolution of faults but also minimizes the dependence on manual labor. This addresses the inherent limitations of traditional methods and rule-based systems, offering a more efficient and reliable approach to network fault management.

V. CONCLUSION

This paper introduces CrossRCA, an intelligent method leveraging LLMs for cross-domain fault localization and RCA in carrier networks. The proposed approach integrates domain-specific knowledge through prompt engineering and RAG, enabling precise fault diagnosis across interdependent network domains, such as OTN and IPRAN. Experimental evaluations on real-world datasets demonstrate the effectiveness of CrossRCA in significantly improving diagnostic accuracy and reducing troubleshooting time, achieving a 90% reduction in traditional fault localization time while providing actionable fault resolution strategies that would otherwise require extensive expert intervention. The results underscore the transformative potential of LLM-driven methodologies to automate fault localization and RCA, enhancing operational efficiency and ensuring robust network performance. Future research will focus on improving the interpretability of the diagnostic process, expanding the applicability of CrossRCA to additional network domains, and incorporating real-time feedback mechanisms to continuously refine its performance.

REFERENCES

- [1] Tiejun J., Xia, Steven Gringeri., Masahito Tomizawa. (2012). High-capacity optical transport networks. *IEEE Communications Magazine*, vol. 50: pp. 170-178.
- [2] Sai Han., Zelin Wang., Guangquan Wang. (2022). Automatic Association of Cross-Domain Network Topology. *IEEE International Conference on Trust, Security and Privacy in Computing and Communications, Wuhan, China*. pp. 1173-1178.
- [3] Yinfang Chen., Huaibing Xie., Minghua Ma. (2024). Automatic root cause analysis via large language models for cloud incidents. In: *Proceedings of the Nineteenth European Conference on Computer Systems*. New York, pp. 674-688.
- [4] Sai Han., D Zhang., Z Wang., G Wang. (2022). Research and applications on intelligent operations of cross-professional carrying network. *Telecommunications Science*. vol. 38: pp. 113-122
- [5] Kaixuan Zhan., Hui Yang., Qiuyan Yao. (2020). Intent Defined Optical Network: Toward Artificial Intelligence-Based Optical Network Automation. In: *Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*. San Diego, USA. pp. 1-3.
- [6] Manish Shetty., Chetan Bansal., Sai Pramod Upadhyayula., Arjun Radhakrishna., Anurag Gupta. (2022). AutoTSG: learning and synthesis for incident troubleshooting. In: *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*. New York, USA, pp. 1477-1488.
- [7] Raouf Boutaba., Mohammad A.. Salahuddin, Noura Limam.. (2018). A Comprehensive Survey on Machine Learning for Networking: Evolution, Applications and Research Opportunities. *Journal of Internet Services and Applications*, pp.1-99.
- [8] Petar Djukic., Maryam Amiri. (2021). Using AI in Network Planning and Operations Forecasting. In: *Proceedings of SCTE Cable-Tec Expo*. pp.1-29.
- [9] Haipeng Yao., Tianle Mai., Xiaobin Xu. (2018). Network AI: An Intelligent Network Architecture for Self-Learning Control Strategies in Software Defined Networks. *IEEE Internet of Things Journal*. vol.5: pp. 4319-4327.
- [10] Stefan Enz., Silvan Nigg., Onur Kalinagac. (2024). ML Driven Root Cause Analysis (RCA) in Telco Microservices Continuum. In: *IEEE International Conference on Communications Workshops*. CO, USA. pp. 371-377.
- [11] Devjeet Roy., Xuchao Zhang., Rashi Bhawe. (2024). Exploring LLM-Based Agents for Root Cause Analysis. In: *Companion Proceedings of the 32nd ACM International Conference on the Foundations of Software Engineering*. New York, USA, pp. 208-219.
- [12] Jialu Zhang., Todd Mytkowicz., Mike Kaufman. (2022). Using pre-trained language models to resolve textual and semantic merge conflicts (experience paper). In: *Proceedings of the 31st ACM SIGSOFT International Symposium on Software Testing and Analysis*. New York, USA. pp. 77-88.
- [13] Pranab Sahoo., Ayush Kumar Singh., Aman Chadha. (2024). A systematic survey of prompt engineering in large language models: Techniques and applications. *ArXiv*. preprint ArXiv:2402.07927.
- [14] Wei Wang., Vincent W. Zheng., Han Yu. (2019). A Survey of Zero-Shot Learning: Settings, Methods, and Applications. *ACM Transactions on Intelligent Systems and Technology*. vol 10: pp.1-37.
- [15] Jason Wei., Xuezhi Wang., Dale Schuurmans. (2024). Chain-of-thought prompting elicits reasoning in large language models. In: *Proceedings of the 36th International Conference on Neural Information Processing Systems*. New York, USA. pp. 24824-24837.
- [16] Patrick Lewis., Ethan Perez., Aleksandra Piktus. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*. vol. 33: pp. 9459-9474.
- [17] Yunfan Gao., Yun Xiong, Xinyu Gao. (2023). Retrieval-Augmented Generation for Large Language Models: A Survey. *ArXiv* abs/2312.10997.
- [18] Shamane Siriwardhana., Rivindu Weerasekera., Elliott Wen. (2023). Improving the domain adaptation of retrieval augmented generation (RAG) models for open domain question answering. *Transactions of the Association for Computational Linguistics*. vol.11: pp. 1-17.
- [19] Zhang, Tianjun., Shishir G. Patil., Naman Jain. (2024). Raft: Adapting language model to domain specific rag. *arXiv* preprint arXiv:2403.10131.